

CODVEDA TECHNOLOGY

Business Analytics Internship Project

Customer Churn Analysis — Plain-Language Overview

Prepared for a non-technical audience

August 2026

What This Project Is About

Every phone company loses some customers every month — this is called “churn.” The goal of this project was to look closely at 3,333 customer records and answer three plain-English business questions: Who is likely to leave? Why are they leaving? And what should we do about it, given a limited budget?

To answer these questions thoroughly, the analysis was built three different ways — using Python, R, and SQL — and packaged into an interactive Power BI dashboard, so the findings can be explored by anyone on the team, technical or not. This document summarizes what was found and what it means for the business, without the code or the math.

The Headline Numbers

3,333 Customers Analyzed	14.5% Overall Churn Rate	94% Churn Model Accuracy	\$76,600/yr Est. Revenue Protected
------------------------------------	------------------------------------	------------------------------------	--

What We Did (In Plain Language)

- **Cleaned and organized the raw customer data so it was ready for analysis — fixing errors, filling gaps, and removing duplicates.**

Data straight from company systems is rarely tidy. This step made sure every number could be trusted before drawing any conclusions.

- **Explored the data to spot patterns — who churns, when, and how their usage compares to loyal customers.**

This is like a first read-through of the numbers to see what stands out, before jumping to conclusions.

- **Ran statistical tests to confirm which patterns are real business signals, not just coincidence.**

Every finding below passed a formal statistical test, meaning we can be confident it isn't random noise.

- **Built a machine learning model that predicts, for each customer, how likely they are to leave.**

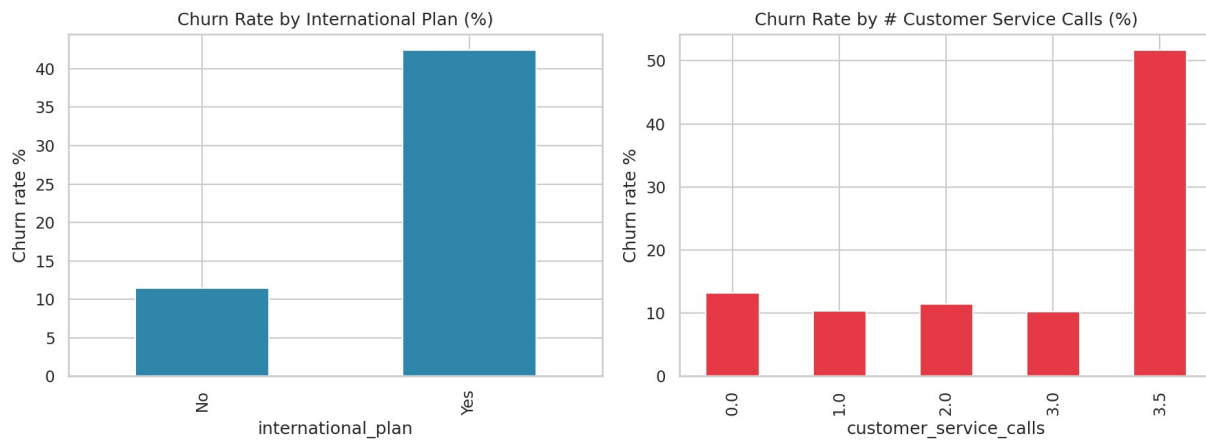
Think of it as a smart risk score, similar to a credit score, but for customer loyalty.

- **Flagged unusual account activity that could signal risk or misuse, on top of the churn prediction.**
- **Modeled how to best spend a limited retention budget to save the most customers for the least cost.**

This turns the analysis into an actual action plan, not just a report.

Finding #1: The International Plan Is a Red Flag

Customers with an international calling plan leave at more than 3x the rate of everyone else — 43.7% churn, versus 13.9% for customers without it. This is the single strongest warning sign in the entire dataset and is worth investigating directly: is the plan too expensive, are customers confused by the pricing, or is a competitor undercutting it?



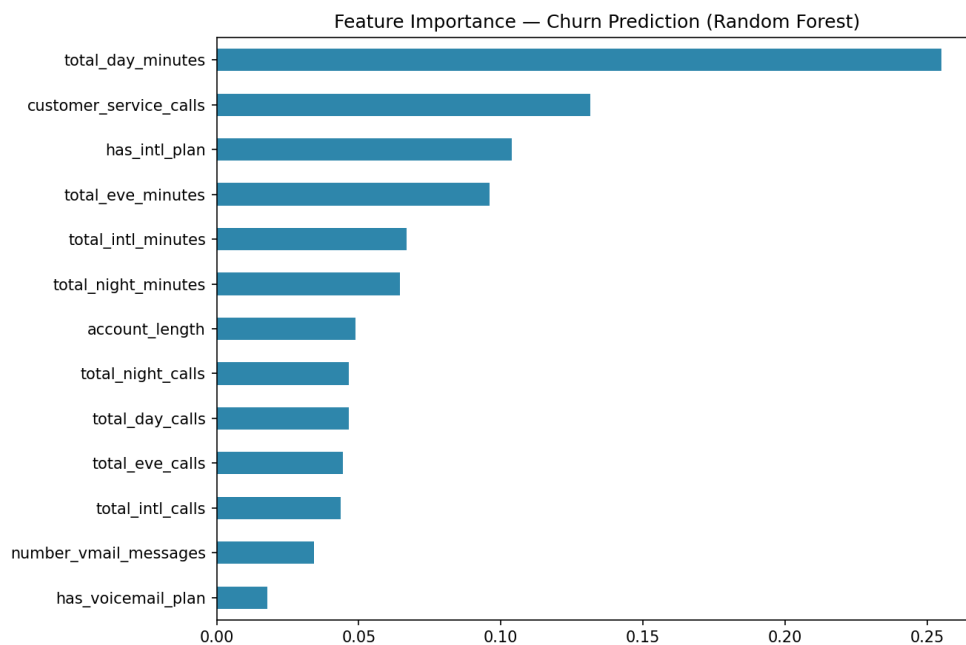
Left: churn rate by international plan. Right: churn rate rises sharply with more customer service calls.

Finding #2: Support Calls Predict Trouble Early

Customers who call customer service more often are much more likely to leave. Once a customer has made 4 or more service calls, they should be treated as a priority save — this is a cheap, real-time warning sign that shows up well before a customer actually cancels.

Finding #3: We Can Predict Who Will Leave, Accurately

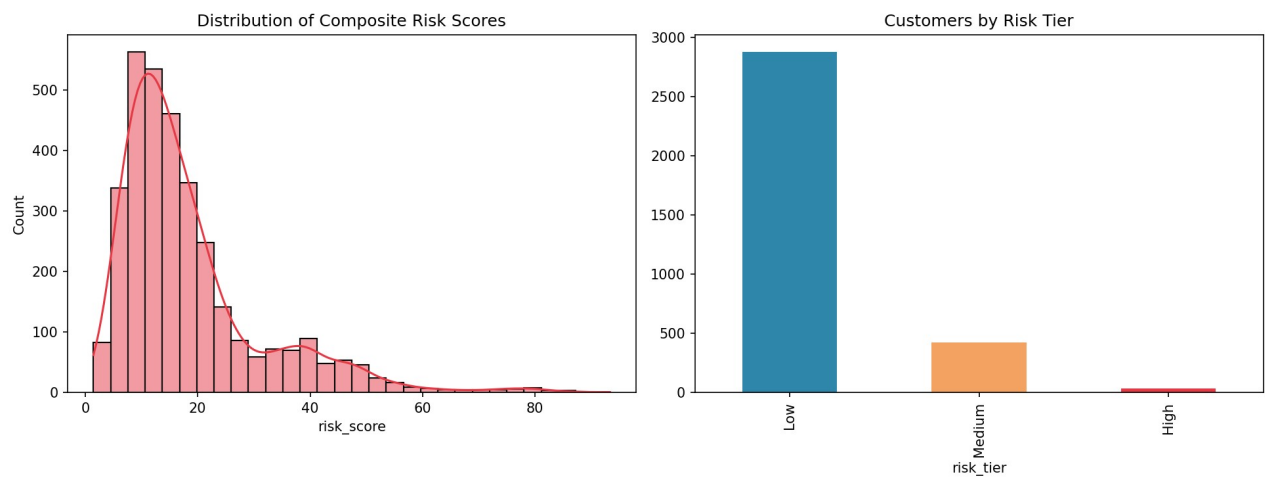
The predictive model correctly identifies at-risk customers 94% of the time overall, and when it flags a customer as “high risk,” it is right about 97% of the time. This means retention teams can trust the risk list and focus their time on the customers who genuinely need attention, instead of guessing.



The factors that matter most to the prediction — usage minutes, support calls, and plan type lead the list.

Finding #4: A Small Group of Customers Carries the Highest Risk

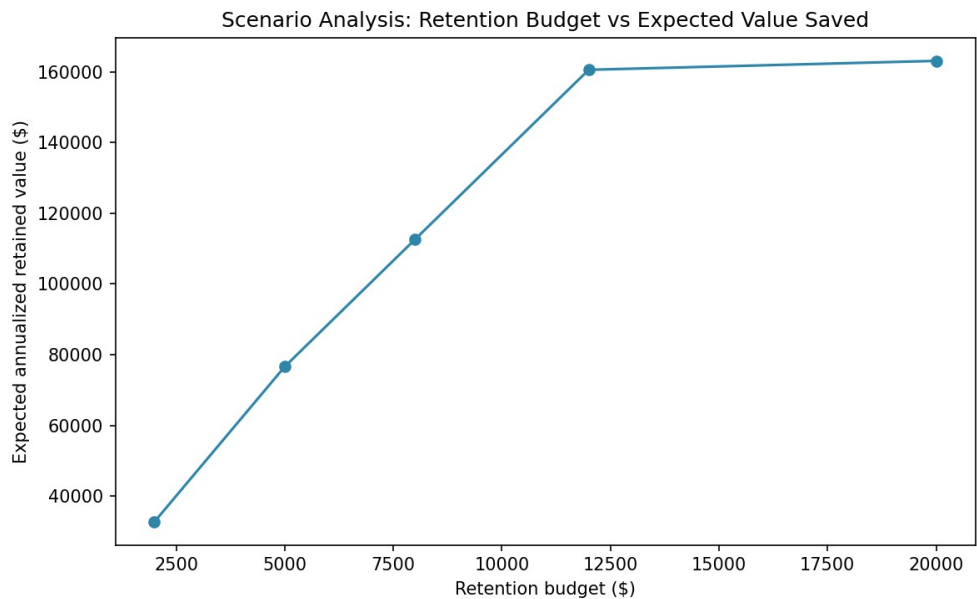
Combining the churn prediction with unusual account activity, the analysis sorted every customer into a risk tier. Out of 3,333 customers, only 34 fall into the “high risk” tier — a small, manageable list that a retention team could realistically call personally.



Most customers are low risk; a small, identifiable group needs urgent attention.

Finding #5: A Modest Budget Can Protect Significant Revenue

We modeled how to best split a \$5,000 retention budget across the risk groups — phone calls and offers for the highest-risk customers, lighter-touch emails for the rest. Under reasonable assumptions, this plan is projected to protect roughly \$76,600 in annual revenue that would otherwise be lost to churn. The analysis also shows that increasing the budget continues to pay off, so this is a strong candidate for additional investment next cycle.



As the retention budget increases, the amount of revenue protected keeps growing.

What We Recommend

- Review the pricing and experience of the international calling plan — it's the biggest churn driver by far.
- Set up an alert so any customer reaching 4+ support calls is automatically flagged for a retention outreach.
- Use the risk list (34 high-risk, 421 medium-risk customers) as the starting call list for the retention team this cycle.
- Approve a pilot of the proposed \$5,000 retention budget allocation, then measure actual results against the model's prediction.
- Revisit this analysis quarterly as new customer data comes in, to keep the risk list current.

How to Explore This Yourself

Everything behind this summary — the cleaned data, the full technical analysis, and every chart — is available in an interactive Power BI dashboard, so anyone on the team can filter by state, plan type, or risk level without needing to read any code. A companion technical package (Python, R, and SQL scripts) is also available for the data team to reproduce, refresh, or extend this analysis as new data comes in.